Predicting Wildfires in California

Using Machine Learning

Alvaro Laham · Ruchi T. Bhanot
Jake Klein · Phillip Dundas
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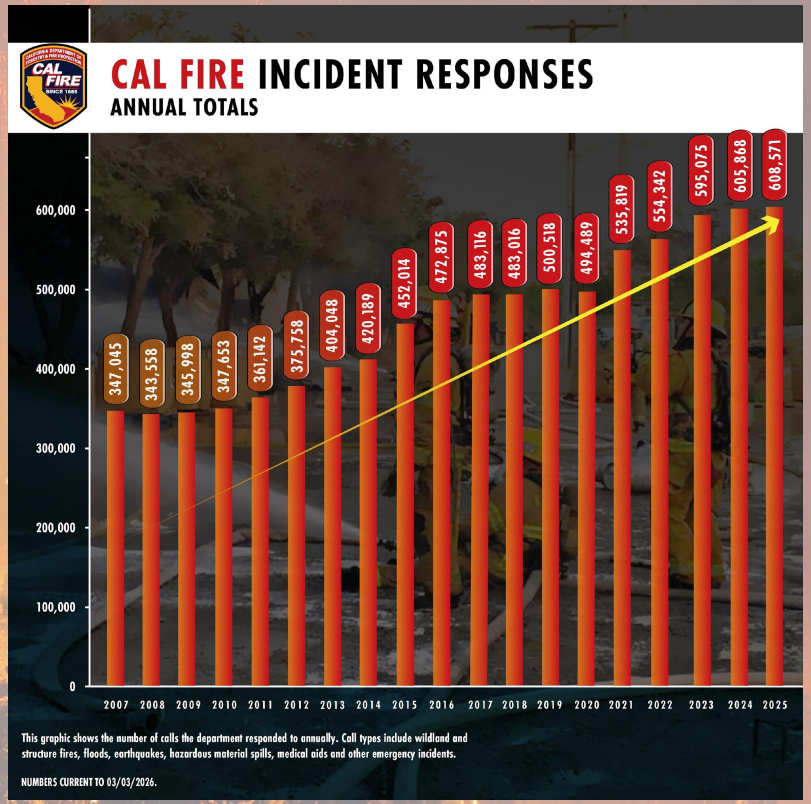
Problem & Motivation

- ***California wildfires have increased dramatically*** in frequency and severity over the past four decades
- Literature review suggests detection systems focused on predicting ***fire expansion*** - too late for proactive resource deployment (Khennou et al., 2023; Radke et al., 2019)
- Weather conditions (temperature, drought, wind, precipitation) are ***known precursors*** to fire ignition (Aurora et al., 2021)
- **Our goal:** predict whether a wildfire will ***start*** on a given day from day-of weather data alone
- **Practical use case:** fire agencies can pre-position crews and equipment on forecasted high-risk days
- We treat this as a binary classification problem — fire day (1) or no-fire day (0)



Our Dataset, Preprocessing & Features

Data source: The California Weather & Fire Prediction Dataset (http://zenodo.org/records/14712845)

14,976	41 yrs	23
Daily Observations	1984 – 2025	Predictive Features

Weather (4)

precipitation · max_temp · min_temp · avg_wind_speed

Engineered (3)

lagged_precipitation · lagged_wind_speed · days_without_rain*

Temporal (16)

Month & season : one-hot encoded

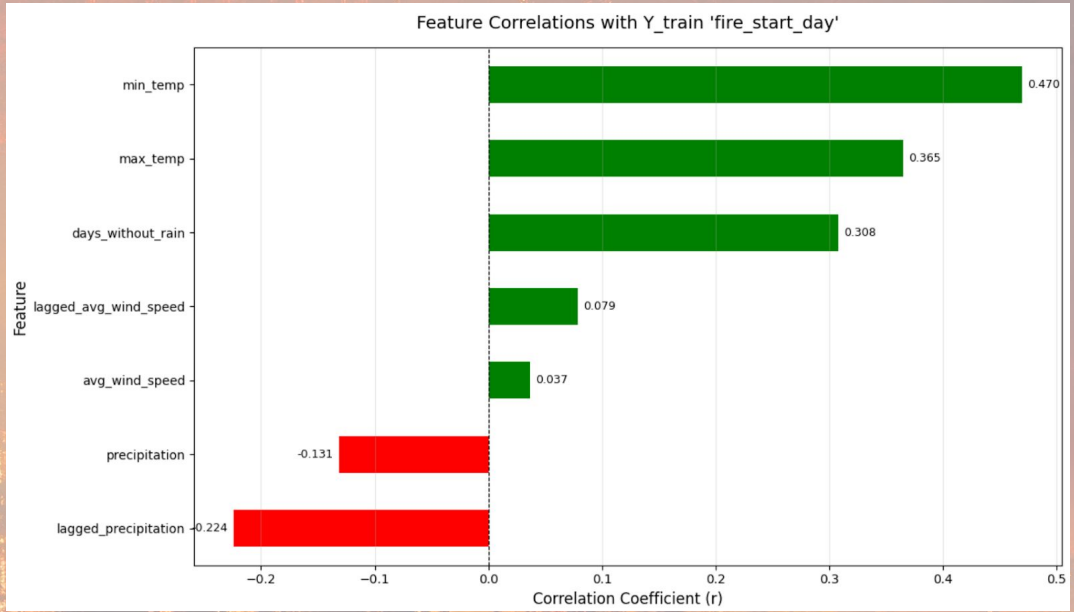
Target (1)

fire_start_day [0 = No Fire | 1 = Fire]

** Team-engineered feature: consecutive days without precipitation*

Exploratory Data Analysis - Key Findings

- **min_temp** has the strongest correlation to fire ignition (**r = 0.47**) - warm overnight temperatures signal elevated risk
- **max_temp** shows strong positive correlation (**r = 0.37**) - daytime heat compounds moisture stress
- **days_without_rain** captures cumulative drought effect (**r = 0.31**) - vegetation dryness builds over time
- **lagged_precipitation -** 7-day prior rainfall - acts as a protective signal (**r = -0.22**)
- **Wind speed** shows very weak correlation with ignition (**r = 0.04 to 0.08**) - more relevant to spread than start



Methodology — Baseline + Four Models

Baseline	Logistic Regression	XGBoost	Random Forest	FFNN
Majority class (always no fire)	Linear boundary TensorFlow	Gradient boosting 1,000-config search	Bagging ensemble Manual grid search	0 to 3 hidden layers Keras Tuner

- **Class imbalance (67/33):** adjusted for by using class_weight='balanced' for Random Forest, scale_pos_weight=2.01 for XGBoost, class weights {0:1, 1:2} for FFNN
- **Feature scaling:** Z-score normalization for logistic regression & FFNN; raw features for XGBoost & Random Forest (tree models are scale-invariant)
- **Tuning:** Keras Tuner Hyperband for logistic regression/FFNN; RandomizedSearchCV (1,000 configs) for XGBoost; manual grid for Random Forest
- **Evaluation:** *F1-score* is primary metric; *recall* is tiebreaker - operationally, missing a fire is worse than a false alarm
- **Train/val/test:** 60/20/20 stratified split; hyperparameters tuned on validation only; test set used once for final evaluation

Results — Model Comparison

Model	Test Accuracy	Fire F1	Fire Recall	Fire Precision
Baseline	66.82%	0.00	0.00	0.00
Logistic Regression	75.47%	68.03%	78.67%	59.92%
XGBoost ★	**76.47%**	**68.79%**	**78.17%**	**61.42%**
Random Forest	75.60%	68.00%	77.00%	60.00%
FFNN	74.50%	67.00%	78.00%	59.00%

Key Takeaways

- XGBoost wins on F1 (68.79%) and highest precision (61.42%) - selected as best model
- All models beat 66.82% baseline by ~8-10% accuracy
- All four models converge near the same F1 (~67-69%)
- Logistic Regression best on recall (78.67%)
- This suggests the bottleneck is the data, not the model.

Seasonal Subgroup Analysis — Random Forest

Season	Count (n)	Fire Days %	Fire Recall	Fire F1
☀️ Summer	791	69.2%	0.995	0.816
🍂 Fall	710	33.0%	0.709	0.552
🌱 Spring	720	22.9%	0.358	0.418
❄️ Winter	775	7.0%	0.000	0.000

Summer: An Illusion of Performance

99.5% recall - but 69% of summer days ARE fire days. Model predicts fire constantly.

Spring: Most Dangerous Gap

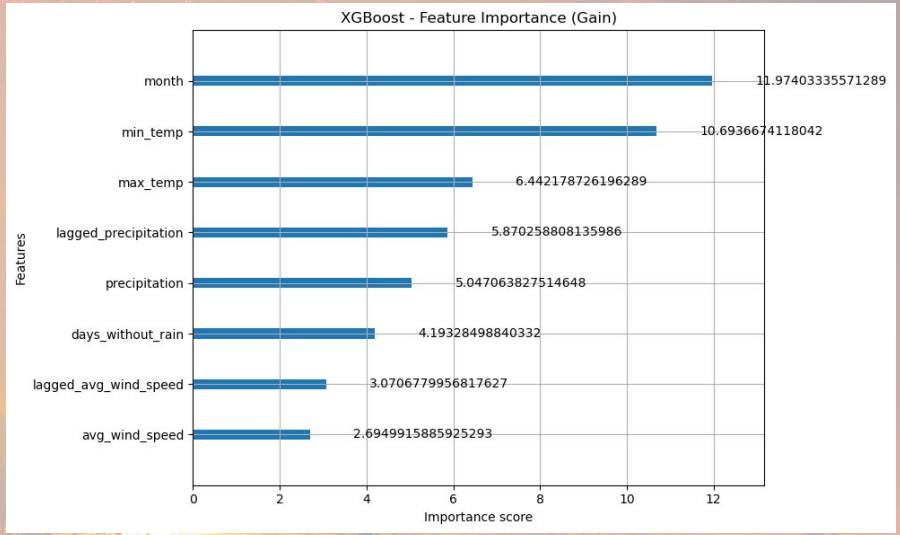
Model misses 64% of spring fires. Low base rate (23%) makes model overly conservative.

Winter: Complete Failure

Zero fires detected. 7% base rate overwhelms class weighting entirely.

Discussion — What the Models Learned

- **Models converge near 76% accuracy across logistic regression, XGBoost, and RF** suggests a data ceiling, not a model complexity ceiling
- **Feature importance: month followed by weather (min_temp, max_temp and lagged participation) to refine prediction -** most efficient way to reduce loss
- **Feature min_temp is the highest-ranked numerical feature**, suggesting fire risk is associated with higher daily low temperatures
- **Geographic blindness: no location identifiers in dataset** - models represent statewide averages, not regional risk
- **Wind not driving ignition:** average and lagged wind speed are not effective at separating data into classes (Fire vs No fire)



Conclusions & Future Work

★ **Best Model: XGBoost | 76.5% Test Accuracy | Fire F1 = 0.69 | Fire Recall = 78.2%**

Key Conclusions

- Monthly and Night time weather data can predict California fire ignition meaningfully above a naive baseline
- All models converged near 76% accuracy - data quality is the limiting factor, not model complexity
- Seasonal patterns are learnable
- Winter and spring fire detection remains an unsolved problem - and challenge - across all models

Future Work

- Add location-specific weather station data to enable regional modeling across California's diverse climate zones
- Incorporate fuel moisture indices and vegetation dryness satellite data as direct physical precursors to ignition
- Collect multi-day weather sequence data to move beyond the IID assumption - fire risk builds over consecutive days and single-day snapshots miss that accumulation
- Investigate season-specific prediction thresholds to improve winter and spring fire detection without sacrificing summer performance

Thank you!

Questions?

References

Aurora A. Gutierrez et al. “Wildfire response to changing daily temperature extremes in California’s Sierra Nevada.” *Science Advances* 7.47 (2021).

Khennou, Fadoua, and Moulay A. Akhloufi. “Improving wildland fire spread prediction using deep U-Nets.” *Science of Remote Sensing* 8 (2023): 100101.

Radke, David, et al. “FireCast: Leveraging deep learning to predict wildfire spread.” *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence (IJCAI-19)* (2019): 4575-4581.